

ROBUST COORDINATION: ELIMINATING THE CONNECTIVITY CONSTRAINT

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! The Fragility of Consensus

Current state of the art methods collapse when networks fragment, causing system wide throughput stalls. These are visible as dark cells in the DMCHBA heatmap at minimum communication range.

🎯 LCBA: Localizing Coordination

The Local Consensus-Based Bundle Algorithm (LCBA) resolves conflicts within each isolated network component, eliminating the need for global connectivity.

📈 Uninterrupted Throughput

LCBA sustains throughput across all arrival rates and communication ranges, matching fully-connected performance even at minimum comm range (4 grid units).

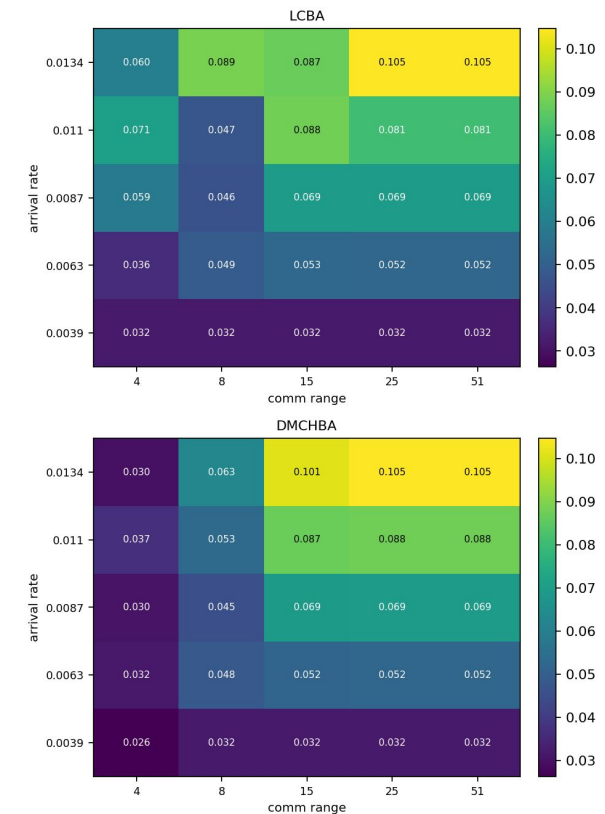


Figure: Steady-State Throughput — Warehouse Small (30×30, 6 agents)